OLYMPIAD · EVERYDAY MATHEMATICS · OLYMPIAD PREPARATION · SHORT NOTES

Chapter 11: Profit, Loss and Discount

Topic-wise short notes with solved examples

Cost price, selling price, profit and loss percentages, marked price and discount, with the shortcuts for successive discounts.

1 Cost price, selling price and profit

- **Cost price (CP)** is what the seller pays. **Selling price (SP)** is what the buyer pays.
- If $\text{SP} > \text{CP}$ there is a **profit**: $\text{profit} = \text{SP} - \text{CP}$.
- If $\text{SP} < \text{CP}$ there is a **loss**: $\text{loss} = \text{CP} - \text{SP}$.
- **Profit percent and loss percent are always calculated on the cost price.**

$$\text{profit}\% = \frac{\text{profit}}{\text{CP}} \times 100 \qquad \text{loss}\% = \frac{\text{loss}}{\text{CP}} \times 100$$

- To find SP from CP: $\text{SP} = \text{CP} \times \left(1 + \frac{\text{profit}\%}{100}\right)$ for a profit, or $\text{CP} \times \left(1 - \frac{\text{loss}\%}{100}\right)$ for a loss.

Example 1.1 A shopkeeper buys a watch for Rs 250 and sells it for Rs 300. Find the profit percent.

SOLUTION

Profit $= 300 - 250 = 50$. Profit percent $= \frac{50}{250} \times 100$.

20%

Example 1.2 A bicycle bought for Rs 400 is sold for Rs 340. Find the loss percent.

SOLUTION

Loss $= 400 - 340 = 60$. Loss percent $= \frac{60}{400} \times 100$.

15%

Example 1.3 Find the selling price needed to gain 20% on an article that costs Rs 450.

SOLUTION

$\text{SP} = 450 \times 1.2$.

Rs 540

2 Finding the cost price from the selling price

- If you know the SP and the profit or loss percent, divide: $\text{CP} = \frac{\text{SP}}{1 \pm \text{percent}/100}$.

Example 2.1 An article is sold at a 10% profit for Rs 1320. Find its cost price.

SOLUTION

$\text{CP} = \dfrac{1320}{1.1}$.

Rs 1200

Example 2.2 An article is sold at a 20% loss for Rs 720. Find its cost price.

SOLUTION

$\text{CP} = \dfrac{720}{0.8}$.

Rs 900

3 Marked price and discount

- The **marked price (MP)** is the price written on the article. A **discount** is a reduction in the marked price.
- $\text{discount} = \text{MP} - \text{SP}$, and $\text{discount}\% = \dfrac{\text{discount}}{\text{MP}} \times 100$. **Discount is calculated on the marked price.**
- Two discounts of $a\%$ and $b\%$ given one after the other are equal to **one** discount of $\left(a + b - \dfrac{ab}{100}\right)\%$.

Example 3.1 A table marked at Rs 800 is sold at a discount of 15%. Find the selling price.

SOLUTION

Discount $= 0.15 \times 800 = 120$.

Rs 680

Example 3.2 Find the single discount equal to two successive discounts of 20% and 10%.

SOLUTION

Factors: $0.8 \times 0.9 = 0.72$. So the buyer pays 72% of the marked price.

28%

Example 3.3 An article costs Rs 500. The shopkeeper marks it 40% above the cost and gives a 10% discount. Find the profit percent.

SOLUTION

$\text{MP} = 500 \times 1.4 = 700$. $\text{SP} = 700 \times 0.9 = 630$.

Profit $= 130$, and $\dfrac{130}{500} \times 100 = 26$.

26%

REMEMBER

Profit and loss use the cost price as the base. Discount uses the marked price as the base. Mixing them up is the most common mistake.

MathSetu